

4.2 The New Study

We divided this section into two parts where we consider two cases of study.

4.2.1 The Study During the Year 2022

In this section, we use the prices of the 24-karat gold during 2022, from the reference [8], to find the mean of the prices for each month in table 4.1. We find the least squares regression line for these data, to predict the average price of the 24-karat gold in January 2023.

	Month	Mean of Prices by SAR
0	January	6813.58 SAR
1	February	6980.39 SAR
2	March	7321.63 SAR
3	April	7263.27 SAR
4	May	6937.15 SAR
5	June	6899.42 SAR
6	July	6533.16 SAR
7	August	6628.29 SAR
8	September	6330.36 SAR
9	October	6262.9 SAR
10	November	6504.13 SAR
11	December	6756.67 SAR

Table 4.1: Gold Ounce Prices from January to December

Now, we will assign each month with one quantity on the x-axis, that is 0 is for January, 1 is for February and so on. From table 4.1, we have the points $(0, 6813.58)$, $(1, 6980.39)$, $(2, 7321.63)$, $(3, 7263.27)$, $(4, 6937.15)$, $(5, 6899.42)$, $(6, 6533.16)$, $(7, 6628.29)$, $(8, 6330.36)$, $(9, 6262.9)$, $(10, 6504.13)$, and $(11, 6756.67)$.

Using the points, we write two matrices X and Y

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \\ 1 & 10 \\ 1 & 11 \end{bmatrix}, Y = \begin{bmatrix} 6813.58 \\ 6980.39 \\ 7321.63 \\ 7263.27 \\ 6937.15 \\ 6899.42 \\ 6533.16 \\ 6628.29 \\ 6330.36 \\ 6262.9 \\ 6504.13 \\ 6756.67 \end{bmatrix},$$

for the regression model $Y = XA + E$, where the coefficients of the least squares regression line are given by the matrix equation,

$$A = (X^T X)^{-1} X^T Y,$$

and E is the error matrix.

Now, we calculate

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \\ 1 & 10 \\ 1 & 11 \end{bmatrix} = \begin{bmatrix} 12 & 66 \\ 66 & 506 \end{bmatrix},$$

and

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{bmatrix} \begin{bmatrix} 6813.58 \\ 6980.39 \\ 7321.63 \\ 7263.27 \\ 6937.15 \\ 6899.42 \\ 6533.16 \\ 6628.29 \\ 6330.36 \\ 6262.9 \\ 6504.13 \\ 6756.67 \end{bmatrix} = \begin{bmatrix} 81230 \\ 437630 \end{bmatrix}.$$

Now, we find the coefficient matrix A as follows.

$$A = (X^T X)^{-1} X^T Y = \begin{bmatrix} 0.2949 & -0.0385 \\ -0.0385 & 0.007 \end{bmatrix} \begin{bmatrix} 81230 \\ 437630 \end{bmatrix} = \begin{bmatrix} 7120.8 \\ -63.9 \end{bmatrix}.$$

So, the least squares regression line is

$$y = 7120.8 - 63.9x.$$

Now, we can use the line to predict the average prices in January 2023. Simply,

we substitute the value 12, which is assigned for January 2023, to get.

$$y = 7120.8 - 63.9(12) = 7057.68.$$

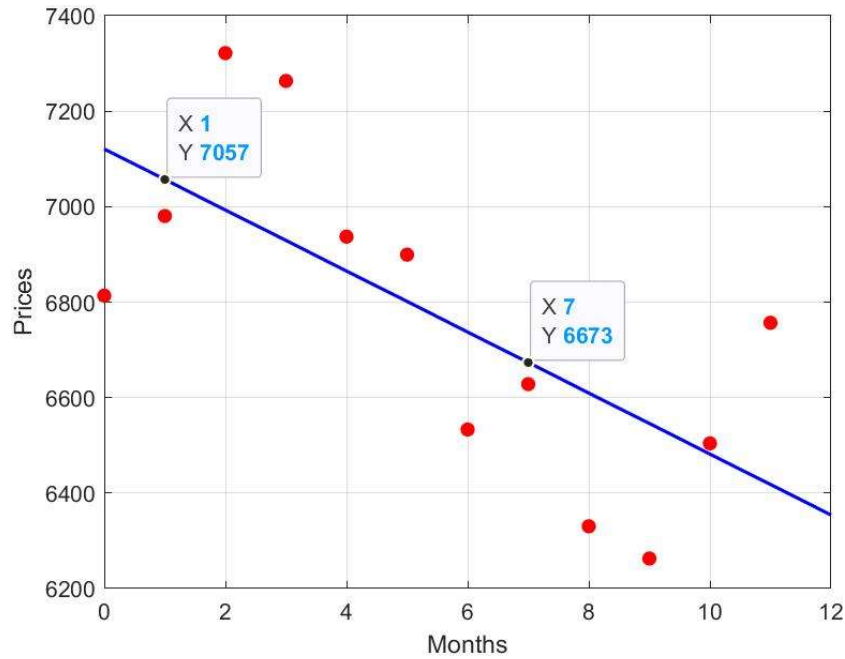


Figure 4.1: The Least Squares Regression Line for Gold Prices During 2022.

Figure 4.1 shows the best line fits the data. To calculate the sum of squared error

$$E^T E,$$

we substitute the values y_i from the table, and the approximate values $f(x_i)$ from the least squares regression line in $e_i = y_i - f(x_i)$, where $i = 1, 2, \dots, 11$.

Then

$$E = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \\ e_{10} \\ e_{11} \end{bmatrix} = \begin{bmatrix} 6813.58 - 7121 \\ 6980.39 - 7057 \\ 7321.63 - 6993 \\ 7263.27 - 6929 \\ 6937.15 - 6865 \\ 6899.42 - 6801 \\ 6533.16 - 6737 \\ 6628.29 - 6673 \\ 6330.36 - 6610 \\ 6262.9 - 6546 \\ 6504.13 - 6482 \\ 6756.67 - 6418 \end{bmatrix} = \begin{bmatrix} -307 \\ -77 \\ 329 \\ 334 \\ 72 \\ 98 \\ -204 \\ -45 \\ -280 \\ -283 \\ 22 \\ 330 \end{bmatrix},$$

and

$$E^T \begin{bmatrix} -307 & -77 & 329 & 334 & 72 & 98 & -204 & -45 & -280 & -283 & 22 & 339 \end{bmatrix}.$$

Thus, the sum of squared error is

$$E^T E = 652298.$$